

2025-12-12 - Servidio (19)

Previous lecture: [2025-12-11 - Servidio \(18\)](#)

Next lecture: [2025-12-18 - Servidio \(20\)](#)

Sweet-Parker model for magnetic reconnection

One of the first model to explain magnetic reconnection

Change of topology

Converts magnetic energy to kinetic energy and heat

1956 Peter Sweet was giving a speech in USA explaining Giovannelli picture of 2 flux tube which create a narrow layer (he was saying "I am working on this before publishig it") and parker (who listened to him) went home and published before Sweet.

Articles:

E.N.Parker, JGR 62, 509, 1957

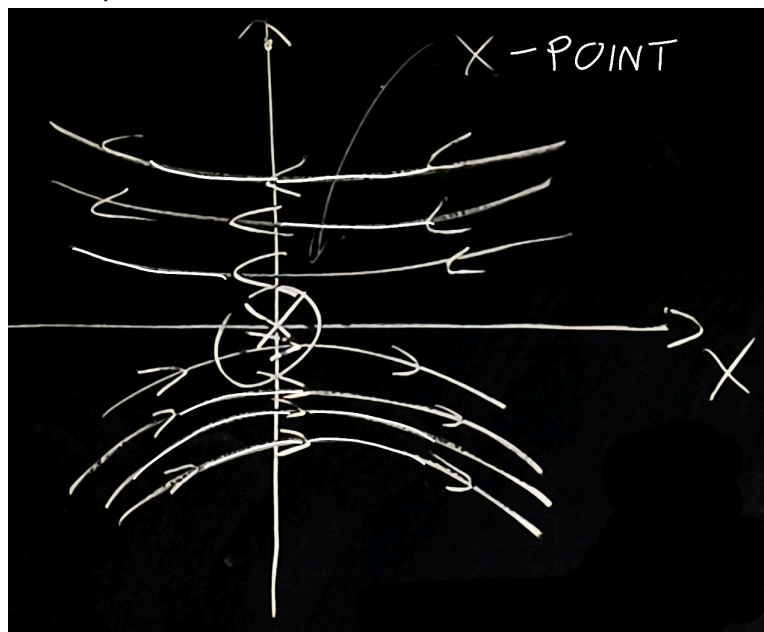
P.A. Sweet, IAU Symposium 6, 123, 1958

Incompressible, resistive MHD

Similar case to 2 flux tube, before we had a line of points with zero magnetic field.

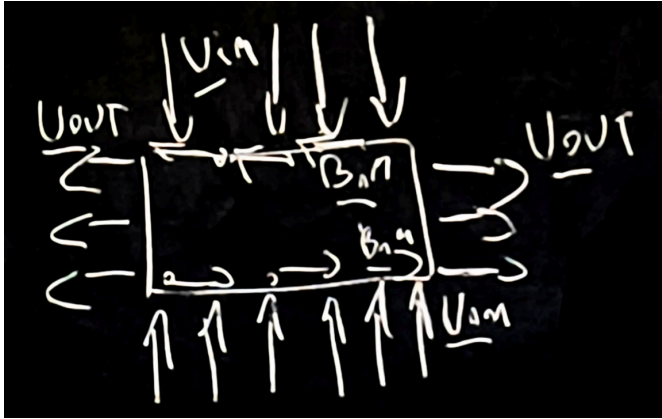
Now we study the case with only one point with zero magnetic field.

PIC X point



We create a black box of side $2L$ along x and 2δ along y .

We are in the steady state



Here we have the perturbation similar to B_0

The important parameters are

$$\bar{U}_{in} = \pm U_{in} \hat{y}; \quad \bar{B}_{in} = \pm B_{in} \hat{x}; \quad \bar{U}_{out} = \pm U_{out} \hat{x}$$

and η, δ, L .

We will look at the reconnection rate U_{in}/U_{out}

The equations are

$$\nabla \cdot \bar{U} = \nabla \cdot \bar{B} = 0$$

$$\frac{\partial \bar{U}}{\partial t} = \bar{U} \times \bar{\omega} - \nabla \left(P + \frac{U^2}{2} + \frac{B^2}{2} \right) + \bar{B} \cdot \nabla \bar{B}$$

$$\frac{\partial \bar{B}}{\partial t} = -\nabla \times \bar{E} = \nabla \times [\bar{U} \times \bar{B} - \eta \bar{J}]$$

We integrate MHD over the black box ($V = 2L \cdot 2\delta$)

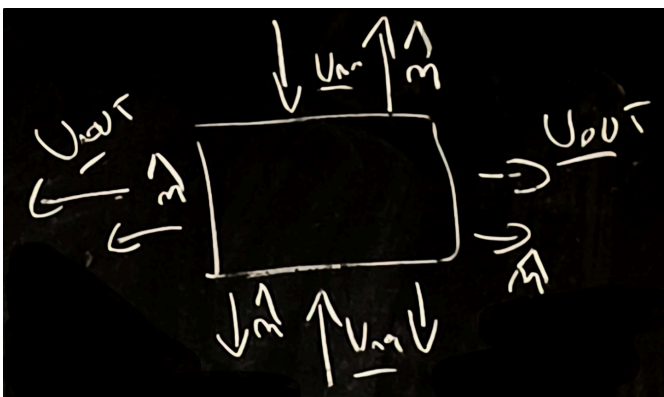
$$\int_V \nabla \cdot \bar{U} d^2x = 0$$

and with $\hat{n}$ outside normal with respect to the box, we have

$$\int_V \nabla \cdot \bar{U} d^2x = \oint \bar{U} \cdot \hat{n} d\ell = \underbrace{-U_{in}2L}_{\text{below}} + \underbrace{U_{out}2\delta}_{\text{left}} - \underbrace{U_{in}2L}_{\text{above}} + \underbrace{U_{out}2\delta}_{\text{right}} = 0$$

where we used the divergence theorem.

PIC box n



The first law is the **conservation of mass**, which reads

$$U_{in}L = U_{out}\delta$$

this is why $U_{out} \gg U_{in}$.

The second law (**conservation of electric field in space**) is derived as follows

$$0 = \frac{\partial \bar{B}}{\partial t} = -\nabla \times \bar{E}$$

So when we move in space, the electric field is conservative.

We take $\bar{E}$ along z because $\bar{U}$ and $\bar{B}$ live in the 2D plane.

$$\bar{E} = -\bar{U} \times \bar{B} + \eta \bar{J} = E_z \hat{z}$$

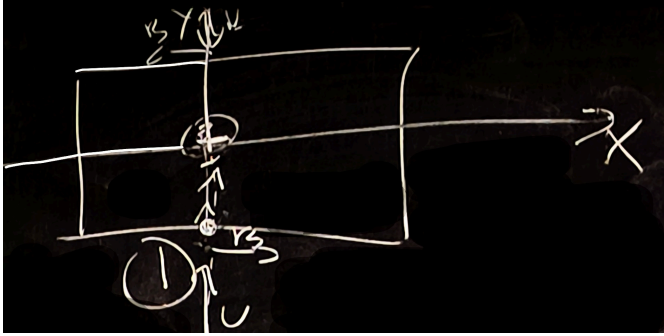
$$\nabla \times (E_z \hat{z}) = 0 \quad \implies \quad \partial_y E_z = \partial_y \left[-\bar{U} \times \bar{B} + \eta \bar{J} \right] = 0$$

which is the second law.

Third law

We want to move from the central-lower part of the box to the X point.

PIC move in the box



From last formula, we get

$$E_1 = E_2 \quad \implies \quad E_{in} = E_X$$

we say that (we neglect J at the border and $B = 0$ in the X point)

$$\begin{cases} E_{in} = -\bar{U}_{in} \times \bar{B}_{in} + \cancel{\eta \bar{J}_{in}} \\ E_X = -\cancel{\bar{U}_X \times \bar{B}_X} + \eta \bar{J}_X \end{cases}$$

so

$$\begin{cases} E_{in} = U_{in} B_{in} \\ E_X = \eta J_X \end{cases} \quad \implies \quad U_{in} B_{in} = E_{in} = E_X = \eta J_X$$

For J_X , we use the rotor of $\bar{B}$

$$J_X = [\nabla \times \bar{B}]_z \simeq \frac{B_{in} - (-B_{in})}{2\delta} = \cancel{2} \frac{B_{in}}{\cancel{2}\delta} = \frac{B_{in}}{\delta}$$

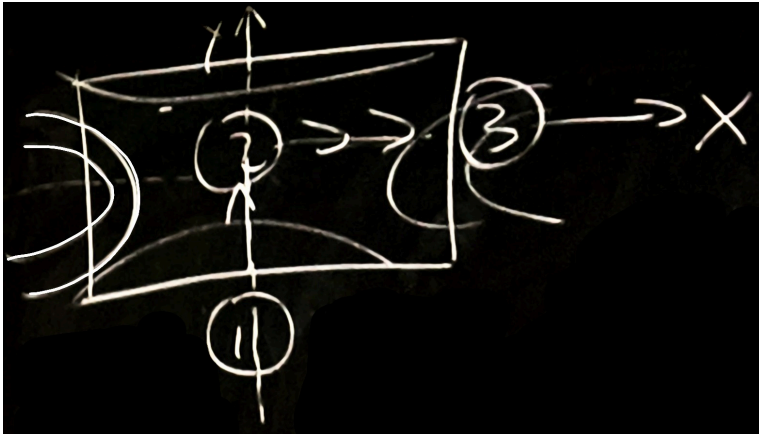
substituting it we have

$$U_{in} B_{in} = \eta J_X = \eta \frac{B_{in}}{\delta} \quad \implies \quad U_{in} = \frac{\eta}{\delta}$$

which is the third law.

Now we move from the X point to the right

PIC formula



We use stationarity

$$\frac{\partial \bar{U}}{\partial t} = 0$$

Moving from 1 to 2 we had

$$0 = \frac{\partial U_y}{\partial t} = (\bar{U} \times \bar{\omega}) \cdot \hat{y} - \partial_y \left[P + \frac{U^2}{2} + \frac{B^2}{2} \right] + \bar{B} \cdot \nabla B_y$$

The first term is null because

$$\bar{U} = U_{in} \hat{y}, \quad \bar{\omega} = \omega \hat{z}$$

The last term is null because

$$\bar{B}_{in} = B_{in} \hat{x} \quad \implies \quad B_y = 0$$

So it must be

$$\partial_y \left[P + \frac{U^2}{2} + \frac{B^2}{2} \right] = 0$$

Now we have the **bernoulli equation for MHD**

$$P + \frac{U^2}{2} + \frac{B^2}{2} = \text{constant}$$

Moving from 2 to 3 ($\bar{U} = U_{out} \hat{x}$, $\bar{\omega} = \omega \hat{y}$, $\bar{B}_{out} \sim 0$)

$$\frac{\partial U_x}{\partial t} = 0 \quad \implies \quad \partial_x \left(P + \frac{U^2}{2} + \frac{B^2}{2} \right) = 0$$

so this is valid everywhere

$$P + \frac{U^2}{2} + \frac{B^2}{2} = \text{constant}$$

in particular

$$P_{in} + \cancel{\frac{U_{in}^2}{2}} + \frac{B_{in}^2}{2} = P_X + \cancel{\frac{U_X^2}{2}} + \cancel{\frac{B_X^2}{2}}$$

we cancel B_X^2 because it's null by definition and U_X^2 because it's stagnant.

We also stay in the hypothesis that $B_{in} \gg U_{in}$.

So the remaining terms yield

$$P_X = P_{in} + \frac{B_{in}^2}{2}$$

Another interesting case (considering $B_{out} \sim 0$) is

$$P_X = P_{out} + \frac{U_{out}^2}{2} + \cancel{\frac{B_{out}^2}{2}} = P_{out} + \frac{U_{out}^2}{2}$$

Comparing the last two results, we have

$$P_{in} + \frac{B_{in}^2}{2} = P_{out} + \frac{U_{out}^2}{2}$$

We say that pressure is constant because gradient of pressure drifts away, so finally

$$\frac{B_{in}^2}{2} = \frac{U_{out}^2}{2}$$

which in Alfvénic notation (and units)

$$B_{in} = U_{out} = U_A$$

which is the again third law (bernoulli theorem).

Summary

So to sum up

$$\begin{cases} U_{in}L = U_{out}\delta & \text{1st law} \\ U_{in} = \frac{\eta}{\delta} & \text{2nd law} \\ U_{out} = U_A & \text{3rd law} \end{cases}$$

From the first law we have

$$\frac{U_{in}}{U_{out}} = \frac{\delta}{L} \implies \delta = L \frac{U_{in}}{U_{out}}$$

From the second and third law, we obtain

$$\frac{U_{in}}{U_{out}} = \frac{\eta}{\delta} \frac{1}{U_A} = \frac{\eta}{\delta U_A}$$

so, using again the first law to substitute the δ , we get

$$\frac{U_{in}}{U_{out}} = \frac{\eta U_{out}}{U_{in} L U_A} = \frac{U_{out}}{U_{in}} \frac{\eta}{L U_A} \implies \frac{U_{in}}{U_{out}} = \sqrt{\frac{\eta}{U_A L}} \equiv S^{-1/2}$$

We define the reconnecting flux as

$$\Psi = \int_{P_0}^{P_x} \bar{\mathbf{B}} \cdot \hat{\mathbf{n}} d\ell$$

PIC reconn flux



with

$$\bar{\mathbf{B}} = \nabla \bar{a} \times \hat{\mathbf{z}} \quad \implies \quad \Psi = a_X - a_0$$

We evaluate the time derivative of the reconnecting flux

$$\frac{d}{dt} \Psi = \frac{d}{dt} a_X - \frac{d}{dt} a_0 = \eta J_X - \cancel{\eta J_0}$$

$$\frac{d}{dt} \Psi = \eta J_X = \frac{\eta U_A}{\delta} = \eta \frac{U_A}{L} \frac{U_{out}}{U_{in}} = \eta \frac{U_A}{L} \sqrt{\frac{U_A L}{\eta}}$$

so

$$\dot{\Psi} = \sqrt{\frac{\eta U_A^3}{L}}$$

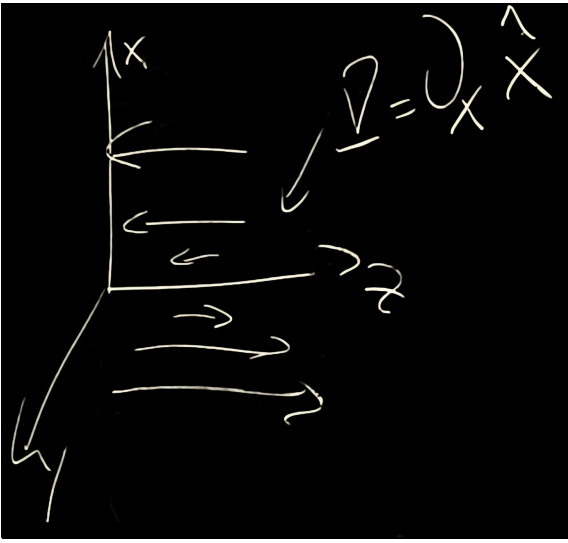
Harris current sheeth (kinetic)

1962

Can we describe a tangential equilibrium in terms of full kinetic theory?

We consider a gradient along x : $\nabla = \partial_x \hat{x}$

PIC xyz



Consider the full Vlasov plasma with no collisions, made up by two species ($\alpha = i, e$)

$$q_i = -q_e = e; \quad n_i \simeq n_e \equiv n$$

We assume stationarity, translational symmetry and non-zero velocities (so we have a current)

$$\frac{\partial}{\partial t} = 0; \quad \partial_y = \partial_z = 0; \quad U_i \neq 0; \quad U_e \neq 0$$

We consider Vlasov for ions and electrons ($\alpha = i, e$)

$$\frac{\partial f_\alpha}{\partial t} + \bar{v} \cdot \nabla f_\alpha + \frac{e}{m_e} \left[\bar{E} + \frac{\bar{V} \times \bar{B}}{c} \right] \cdot \nabla_{\bar{v}} f_\alpha = 0$$

which is coupled with Maxwell equations

$$\begin{cases} \nabla \cdot \bar{E} = 4\pi\rho_{tot} \\ \nabla \times \bar{E} = -\frac{1}{c} \frac{\partial \bar{B}}{\partial t} \\ \nabla \cdot \bar{B} = 0 \\ \nabla \times \bar{B} = \frac{4\pi}{c} \bar{J} + \frac{1}{c} \frac{\partial \bar{E}}{\partial t} \end{cases}$$

In stationarity conditions ($\partial/\partial t = 0$), Maxwell equations gets reduced to

$$\begin{cases} \nabla \cdot \bar{E} = 4\pi\rho_{tot} = 4\pi e [n_i + n_e] \\ \nabla \times \bar{B} = \frac{4\pi}{c} \bar{J} = \frac{4\pi}{c} [en_i \bar{U}_i - en_e \bar{U}_e] \end{cases}$$

or, written in terms of the distribution function

$$\begin{cases} \nabla \cdot \bar{E} = 4\pi e \left[\int f_i d^3v - \int f_e d^3v \right] \\ \nabla \times \bar{B} = \frac{4\pi}{c} e \left[\int \bar{v} f_i d^3v - \int \bar{v} f_e d^3v \right] \end{cases}$$

with $f_\alpha(x, v_x, v_y, v_z)$ discontinuous.

Other supporting equations are (using $\partial_y = \partial_z = 0$)

$$\bar{B} = \nabla \times \bar{A} = \frac{\partial A_y}{\partial x} \cdot \hat{z} \quad \Longrightarrow \quad \nabla \times \bar{B} = -\frac{\partial^2 A_y}{\partial x^2} \cdot \hat{y}$$

and

$$\nabla \times \bar{E} = \frac{\partial \bar{B}}{\partial t} = 0 \quad \Longrightarrow \quad \bar{E} = -\nabla \phi \quad \Longrightarrow \quad \nabla \cdot \bar{E} = -\nabla^2 \phi = -\frac{\partial^2 \phi}{\partial x^2}$$

so these two equations become the Poisson equation for the electric potential and a simplified equation to find the vector potential

(we multiply $\nabla \times \bar{B}$ by $\hat{y}$ because other components are null)

$$\begin{cases} \frac{\partial^2 \phi}{\partial x^2} = -4\pi e \left[\int f_i d^3v - \int f_e d^3v \right] \\ \frac{\partial^2 A_y}{\partial x^2} = -\frac{4\pi}{c} e \left[\int \bar{v} f_i d^3v - \int \bar{v} f_e d^3v \right] \cdot \hat{y} \end{cases}$$

Previous lecture: [2025-12-11 - Servidio \(18\)](#)

Next lecture: [2025-12-18 - Servidio \(20\)](#)